# CS3210 Parallel Computing

AY 25/26 Sem 2 — github/omgeta

## 1. Introduction

Parallel Computing is the simultaneous use of multiple processing units to solve problems efficiently.

Program Parallelization Steps:

i. Decomposition: split problem into parallel tasks
   - Granularity: size of task

ii. Scheduling: assign tasks to processes/threads
   - Orchestration: imposed synchronisation and communication of tasks to satisfy dependencies

iii. Mapping: bind processes/threads to hardware processing units (e.g. CPU cores)

### Processes

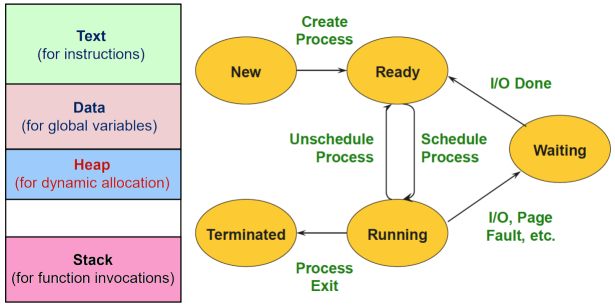
Process is an abstraction for a running program:

i. Process ID (PID): uniquely identifies a process

ii. Process State: indicates execution status

iii. Process Control Block: stores execution context (registers, resources, private address space)

– Costly: syscall overhead, data structures must be allocated, communication goes through OS




Interprocess Communication (IPC):

i. Shared Memory (e.g. locks, semaphores)

ii. Message Passing

iii. UNIX Pipes and Signals

### Threads

Threads are independent execution flows within a process:

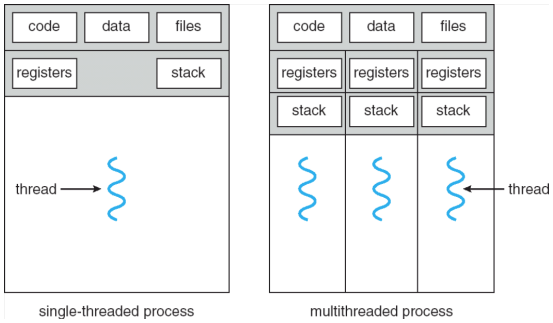
i. Shared: Resources, Address Space

ii. Private: Thread ID, Registers, "Stack" (diff. SP)

\+ Efficient: cheaper creation and context switching

\+ Resource Sharing: process resources can be shared

\+ Concurrent: multithreading on different cores




Implementations:

i. User-Threads are managed by a userspace library:
   + Fast: context switches have low overhead
   – No Parallelism: OS cannot map threads to different execution units
   – Threads blocking will block whole process

ii. Kernel-Threads are implemented in the OS:
   + Parallel: threads can run across many CPUs
   – Slower: thread operations are system calls

Mapping Models:

i. Many-to-One:
   - All user-level threads are mapped to one process
   - User library schedules user-threads

ii. One-to-One:
   - Each user-thread mapped to one kernel-thread
   - No user library scheduler needed

iii. Many-to-Many:
   - User-threads mapped to set of kernel-threads
   - Library scheduler may move user-threads to different kernel-threads

### Synchronisation

Synchronisation restricts interleavings of concurrent threads to coordinate access to shared resources.

Critical Section protects resources from concurrent access:

i. Safety: nothing bad happens
   - Mutual Exclusion: at most 1 thread in CS

ii. Liveness: something good happens
   - Progress: if no thread is in CS, a waiting thread should be granted access
   - Bounded Wait: a waiting thread requesting to enter the CS, will eventually enter

iii. Performance: overhead is small w.r.t work done

Symptoms of Incorrect Synchronization:

i. Deadlock: all threads waiting; iff all conditions met
   - Mutual Exclusion: $\geq 1$ resource held exclusively
   - Hold & Wait: $\geq 1$ process holding one resource while waiting for another
   - No Pre-emption: resources must be yielded
   - Circular Wait: set of processes waiting in circles

ii. Starvation: a thread is unable to make progress
   - Livelock: due to deadlock avoidance

iii. Race Condition: bug from program outcome depending on unpredictable access order

iv. Data Race: two threads access a shared resource with no synchronisation, and one is a write

Mechanisms:

i. Lock: primitive `acquire()`, `release()`
   - Spinlock (busy-wait) vs. Mutex (blocking)
   - Uses hardware atomics (`TSL`) or toggle interrupts

ii. Semaphore: atomic counter $\geq 0$; `wait()`, `signal()`
   - Binary/Mutex: represents single access
   - Counting/General: number of threads in CS $\leq N$
   - Can be manipulated by different thread
   - No relation to data being controlled

iii. Monitor: high-level implicit sync data structure

iv. Messages

## Producer-Consumer

Processes share a buffer, where producers add items and consumers remove items when not empty.

Unbounded Buffer (`mutex = Sem(1)`, `items = Sem(0)`)

```
// Producer
x = produce();
mutex.wait();
  buffer.add(x);
mutex.signal();
items.signal();
```

```
// Consumer
items.wait();
mutex.wait();
  x = buffer.get();
mutex.signal();
consume(x);
```

Finite Buffer (`spaces = Sem(K)`):

```
// Producer
x = produce();
spaces.wait();
mutex.wait();
  buffer.add(x);
mutex.signal();
items.signal();
```

```
// Consumer
items.wait();
mutex.wait();
  x = buffer.get();
mutex.signal();
spaces.signal();
consume(x);
```

## Lightswitch

```
class Lightswitch:
  def __init__(self):
    self.counter = 0
    self.mutex = Sem(1)

  def lock(self, sem):
    self.mutex.wait()
      self.counter++
      if self.counter == 1:
        sem.wait()
    self.mutex.signal()

  def unlock(self, sem):
    self.mutex.wait()
      self.counter--
      if self.counter == 0:
        sem.signal()
    self.mutex.signal()
```

## Barrier

```
bool last = false;
mutex.wait();
count++;
if (count == N) {
  last = true;
  barrier.signal();
}
mutex.signal();

barrier.wait();
if (!last)
  barrier.signal();
```

## Reader-Writer

Processes share a critical region, where readers can read simultaneously but writers must have exclusive access.

```
// Writer
empty.wait();
  write();
empty.signal();
```

```
// Reader
mutex.wait();
  readers++;
  if (readers == 1)
    empty.wait();
mutex.signal()
read();
mutex.wait();
  readers--;
  if (readers == 0)
    empty.signal();
mutex.signal()
```

Lightswitch:

```
// Writer
empty.wait();
  write();
empty.signal();
```

```
// Reader
rswitch.lock(empty);
  read();
rswitch.unlock(empty);
```

No Starvation (`gate = Sem(1)`):

```
// Writer
gate.wait();
empty.wait();
  write();
gate.signal();
empty.signal();
```

```
// Reader
gate.wait();
gate.signal();
rswitch.lock(empty);
  read();
rswitch.unlock(empty);
```

Priorities:

```
// Writer
wswitch.lock(noRead);
noWrite.wait();
  write();
noWrite.signal();
wswitch.unlock(noRead);
```

```
// Reader
noRead.wait();
rswitch.lock(noWrite);
noRead.signal();
  read();
rswitch.unlock(noWrite);
```

## Dining Philosophers

$N$ philosophers around a circular table, with a single chopstick between. 2 chopsticks are needed to eat.

```
// Philosopher
void p(int i) {
  while (1) {
    // think
    take(i);
    eat();
    put(i);
  }
}

void safe_to_eat
    (int i) {
  if (state[i] ==
   HUNGRY &&
   state[LEFT] !=
   EATING &&
   state[RIGHT] !=
   EATING) {
    state[i] =
   EATING;
    signal(s[i]);
  }
}
```

```
// Take Chopstick
void take(int i) {
  wait(mutex);
  state[i] =
   HUNGRY;
  safe_to_eat(i);
  signal(mutex);
  wait(s[i]);
}

// Put Chopstick
void put(int i) {
  wait(mutex);
  state[i] =
   THINKING;

   safe_to_eat(LEFT);

   safe_to_eat(RIGHT)
  signal(mutex);
}
```

Limited Eater Solution, when at most $N-1$ philosophers eat concurrently, it's guaranteed at least one can eat:

```
void philosopher(int i) {
  while (1) {
    // think
    wait(seats);
    wait(chopstick[LEFT]);
    wait(chopstick[RIGHT]);
    eat();
    signal(chopstick[RIGHT]);
    signal(chopstick[LEFT]);
    signal(seats);
  }
}
```

## 2. Architecture

Levels of Parallelism:

i. Bit-Level: operate on wider words
   - SIMD: operate on superword/vector registers

ii. Instruction-Level:
   - Pipeline: split execution into stages; multiple instructions can occupy different stages in same cycle (if no dependency hazards)
   - Superscalar: duplicate pipelines; issue multiple independent instructions per cycle

iii. Thread-Level: execute multiple threads on the same core in parallel (i.e. SMT)

iv. Processor-Level: execute multiple threads/processes in parallel on multiple cores/processors

Flynn Taxonomy of Parallel Architectures:

i. Single Instruction Single Data (SISD): single stream, each instruction on single data

ii. Single Instruction Multiple Data (SIMD): single stream, each instruction on multiple data

iii. Multiple Instruction Single Data (MISD): multiple streams, each instruction on same data

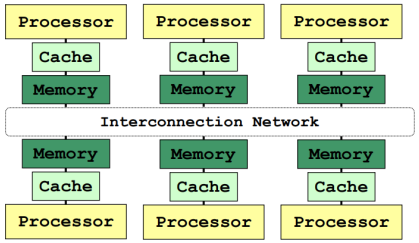
iv. Multiple Instruction Multiple Data (MIMD): per-PU stream to work on its own data

### Distributed-Memory

Distributed-Memory Systems are multicomputer systems:

i. Node: indep. unit of processor and private memory

ii. Interconnection Network: supports message passing for data exchange between nodes




### Shared-Memory

Shared-Memory Systems are multiprocessor systems:

i. Shared Memory Provider: maintains shared address space abstraction across PUs

\+ No need to partition code or data

\+ Efficient communication: no need to physically move data between processors

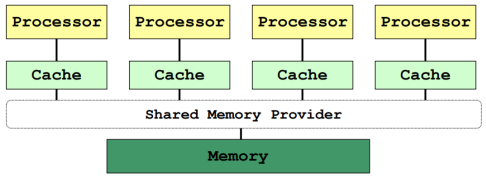
– Special synchonisation constructs required

– Limited scalability due to contention

Uniform Memory Access (UMA):

i. Same memory access latency for every PU

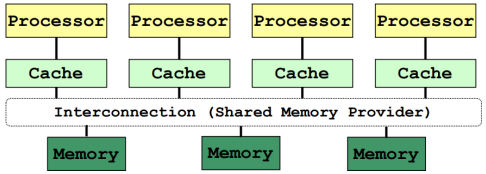
\+ Suitable with few processors due to contention




Non-Uniform Memory Access (NUMA):

i. Different memory access latency for every PU (e.g. local memory faster than remote)

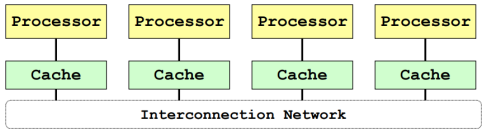
ii. Cache Coherent NUMA (ccNUMA): each node has coherent cache to reduce contention




Cache Only Memory Access (COMA):

i. Memory blocks behave as caches

ii. Data migrates dynamically and continuously according to cache coherence scheme




### Cache Coherence

Caches reduce memory access latency and provide high bandwidth data transfer to CPU:

i. Cache Size: larger cache reduces misses but increases access time (addressing complexity)

ii. Block Size: unit of transfer; larger blocks improve spatial locality, but longer transfer time

Write Policy:

i. Write-through: write to both cache and memory
   \+ Always get newest value of memory block
   – Slower due to many memory accesses; mitigate with write buffer

ii. Write-back: write to cache only and set dirty bit; write to memory on cache block eviction
   \+ Fewer write operations to memory
   – Memory may contain invalid entries

Coherence defines behaviour of reads/writes to the same memory location. Properties:

i. Program Order: if $P$ writes to $X$, n.f.w to $X$, $P$ should read same value from $X$

ii. Write Propagation: if $P_1$ writes to $X$, n.f.w to $X$, $P_2$ should read same value from $X$ eventually

iii. Transaction Serialization: if $V_1 \to X$, then $V_2 \to X$, all $P_i$ should never read $V_2$ then $V_1$ from $X$

Implementations:

i. Software-based: use page-fault to propagate writes

ii. Snooping-based: cache controllers snoop/broadcast on shared bus

iii. Directory-based: central directory records sharing status; no broadcast; common in NUMA

Implications:

i. "Increase" memory latency and lower cache hit rate

ii. Cache Ping-Pong: 2 PUs write to same cache line; invalidations slow fetch from memory

iii. False Sharing: PUs write to different addresses on same cache line; avoid with padding

## Memory Consistency

Memory Consistency constrains the order in which memory operations by a thread become visible (commit) to other threads for different memory locations.

Memory Ordering Requirements:

i. $W \to R$: write to $X$ commits before read from $Y$
ii. $R \to R$: read from $X$ commits before read from $Y$
iii. $R \to W$: read from $X$ commits before write to $Y$
iv. $W \to W$: write to $X$ commits before write to $Y$

Sequential Consistency Model (SC):

i. Preserves all four ordering requirements
ii. PUs issue memory operations in program order
iii. Interleaving of PUs' program orders is acceptable
– Can result in loss of performance

Relaxed Consistency Model:

i. Relax ordering only if no data dependencies between operations to same memory location:
   - $R \to W$: anti-dependence (WAR)
   - $W \to W$: output dependence (WAW)
   - $W \to R$: flow dependence (RAW)
ii. Override constructs can prevent certain re-orderings
iii. Total Store Ordering (TSO):
   - Relax $W \to R$: a PU may read $B$ before its earlier write to $A$ is visible to *all* other PUs
   - Write Atomicity: writes become visible to all PUs at the same time
iv. Processor Consistency (PC):
   - Relax $W \to R$
   - No Write Atomicity: a write may be seen by some PUs before others
v. Partial Store Ordering (PSO):
   - Relax $W \to R$
   - Write Atomicity
   - Relax $W \to W$: a later write to $B$ can bypass an earlier write to $A$

\+ Hide latencies by overlapping indep. operations

## Interconnection Networks

Interconnection forms the backbone of communication between processors, memories/caches, and I/O devices.

## Topology

Topology is the geometrical shape of the connection.

Direct Interconnections (or Static, P2P) have endpoints connected directly, usually of same type (e.g. core–core). Metrics:

i. Diameter, $\delta(G)$: maximum shortest-path distance between any pair of nodes
   - Small diameter $\Rightarrow$ shorter message transmission
ii. Degree, $g(v)$: no. of nodes adjacent to node $v$; $g(G)$ is the maximum node degree in network $G$
   - Small degree $\Rightarrow$ lower node hardware overhead.
iii. Bisection Width, $B(G)$: minimum no. of edges removed to split the network into two equal halves
   - Bisection bandwidth $BW(G)$: total bandwidth between the two bisections
   - Measures capacity under simult. transmission
iv. Node Connectivity, $nc(G)$: minimum no. of nodes whose failure disconnects the network
   - Determines robustness of the network
v. Edge Connectivity, $ec(G)$: minimum no. of edges whose failure disconnects the network
   - Determines number of independent paths between any pair of nodes

| | $n$ | $g(G)$ | $\delta(G)$ | $ec(G)$ | $B(G)$ |
|---|---|---|---|---|---|
| Complete Graph | $n$ | $n-1$ | 1 | $n-1$ | $(\frac{n}{2})^2$ |
| Linear Array | $n$ | 2 | $n-1$ | 1 | 1 |
| Ring | $n$ | 2 | $\lfloor \frac{n}{2} \rfloor$ | 2 | 2 |
| $d$-dim Mesh | $r^d$ | $2d$ | $d(\sqrt[d]{n}-1)$ | $d$ | $n^{\frac{d-1}{d}}$ |
| $d$-dim Torus | $r^d$ | $2d$ | $d\lfloor \frac{\sqrt[d]{n}}{2} \rfloor$ | $2d$ | $2n^{\frac{d-1}{d}}$ |
| $k$-dim Hypercube | $2^k$ | $\log n$ | $\log n$ | $\log n$ | $\frac{n}{2}$ |
| $k$-dim CCC | $k2^k$ | 3 | $2k-1+\lfloor \frac{k}{2} \rfloor$ | 3 | $\frac{n}{2k}$ |
| Complete BT | $2^k-1$ | 3 | $2\log \frac{n+1}{2}$ | 1 | 1 |
| $k$-ary $d$-cube | $k^d$ | $2d$ | $d\lfloor \frac{k}{2} \rfloor$ | $2d$ | $2k^{d-1}$ |

Indirect Interconnection (or Dynamic) have endpoints interconnect through switches configured dynamically:

\+ Reduce hardware costs by sharing switches & links

Metrics:

i. Cost: number of switches and links
ii. Concurrent connections

Types:

i. Bus: set of wires carry data from sender to receiver
   - Only a pair of devices can communicate at a time
   - Bus arbiter used for coordination
   - Typically used for small number of processors
ii. Crossbar: switch matrix (of $n \times m$ switches) from $n$ inputs to $m$ outputs
   - Switch states: straight or direction change
   - Hardware costly $\to$ small number of processors
iii. Multistage Switching: intermediate switches with connecting wires between neighbouring stages
   - Goal: obtain small distance for any pair of input and output devices
iv. Omega: unique path from each input to each output ($n \times n$)
   - Construction: switch $(\alpha, i) \to (\beta, i+1)$ where $\beta$ is cyclic-left-shift of $\alpha$, and cyclic-left-shift + inversion of LSB
   - $\log n$ stages of $\frac{n}{2}$ switches per stage
v. Butterfly:
   - Construction: switch $(\alpha, i) \to (\alpha, i+1), (\alpha', i+1)$ where $\alpha, \alpha'$ differ in the $(i+1)^{th}$ bit from the left
vi. Baseline:
   - Construction: switch $(\alpha, i) \to (\beta, i+1)$ where $\beta$ is cyclic-right-shift of last $(k-i)$ bits of $\alpha$, and inversion of LSB + same cyclic-right-shift

## Routing

Routing determines path(s) from source to destination within a given topology. Algorithm Classification:

i. Path Length:
   - Minimal: shortest-path always chosen
   - Non-minimal: shortest-path not necessary

ii. Adaptivity:
   - Deterministic: always use same path for same pair of source and destination nodes
   - Adaptive: take into account network status (e.g. avoid congested path, avoid dead nodes)

Deterministic Algorithms:

i. XY Routing for 2D Mesh:
   - Move in $X$ direction until $X_{src} == X_{dst}$
   - Move in $Y$ direction until $Y_{src} == Y_{dst}$

ii. E-Cube Routing for Hypercube:
   - Bit difference in source and target node address (hamming distance) = number of hops
   - From MSB→LSB (or vice versa), find first different bit and go to neighbour node with bit corrected; at most $n$ hops

iii. XOR-Tag Routing for Omega Network:
   - Let $T$ = Source ID $\oplus$ Destination ID
   - At stage $k$, go straight if bit $k = 0$, else crossover

## Additional

More Questions:

i. Switching: how to transfer messages along a path

ii. Flow Control: how to handle concurrent messaging

## Current Trends

Ethernet: supports P2P and broadcast; common topologies include bus, and star with a logical bus.

InfiniBand: supports P2P and multicast; common topologies include fat-tree and torus.

# 3. Programming Models

Types of Parallelism:

i. Data Parallelism: split data among PUs (loop parallelism; SIMD/SPMD)

ii. Task Parallelism: split indep. tasks among PUs (statements, loops, function calls)
   - Task Dependency Graph: DAG representing control dependency between tasks
     - Critical Path Length: slowest completion
     - Deg. Concurrency: $\dfrac{\text{Total Work}}{\text{Critical Path Length}}$

Representation of Parallelism:

i. Implicit Parallelism: Haskell

ii. Implicit Scheduling: OpenMP

iii. Explicit Communication & Sync: MPI, Pthreads

Models of Coordination:

i. Shared Address Space: very little structure
   - All threads can access all shared variables; mutual exclusion via locks
   - Match shared-memory systems (UMA/NUMA); requires HW support; costly to scale
   - \+ Simple to program, if synced correctly
   - − Scalability limited by memory cost variation (even with NUMA) or bandwidth limiting speed

ii. Message Passing: highly structured
   - Tasks operate in private address spaces; communication via explicit send/receive
   - Matches distributed memory systems
   - \+ Allows scaling across many machines
   - − Complex programming
   - − Network communication delays

## Foster's Design Methodology

i. Partitioning: problem into pieces/data
   - $\geq 10\times$ primitive tasks than cores, of similar size
   - Minimize redundant computations and data
   - Number of tasks increases with problem size

ii. Communication: pass needed data among tasks
   - Balance communication among tasks
   - Perform communication in parallel
   - Overlap computation with communication
   - Communicate with small number of neighbours

iii. Agglomeration: combine tasks into larger tasks
   - Number of tasks $\geq$ number of cores
   - Reduce cost of task creation + communication
   - Maintain scalability, simplify code

iv. Mapping: tasks to execution units
   - Maximize processor utilization; minimize IPC
   - Optimal mapping is NP-hard $\Rightarrow$ heuristics
   - Consider static vs dynamic allocation; if static, tasks:cores $\geq$ 10:1; if dynamic, allocator must not bottleneck

## Parallel Programming Patterns

Patterns:

i. Fork–Join: `fork` child tasks, then `join` (D&C)

ii. Parbegin–Parend: mark statements to execute in parallel threads, then sync at end

iii. SIMD: single instruction stream over multiple data elements (synchronous)

iv. SPMD: same program on multiple threads; different data via IDs (asynchronous)

v. Master–Worker: master distributes tasks to workers

vi. Task Pools: shared pool of tasks; fixed number of worker threads dynamically dequeue/insert tasks

vii. Producer–Consumer: produce tasks into buffer for consumers to retrieve; sync coordinates passing

viii. Pipelining: split compute into stages for data stream; throughput improves via stage-overlap

## OpenMP

Shared-Memory Programming Model:

i. Threads communicate via shared variables; may also have private variables
ii. Avoid race conditions using mutual exclusion

### Constructs

General:

i. `#pragma omp parallel`: define parallel region (fork–join), creating team of threads in the block
ii. `omp_get_thread_num()`/`omp_get_num_threads()`: query a thread's id and the current team size.
iii. `omp_set_num_threads(n)` / `OMP_NUM_THREADS`: control the number of threads
iv. `nowait`: disable implicit barrier after blocks

Data Parallel For:

i. `#pragma omp for`: work-sharing for loops
ii. `schedule(kind,chunksize)`: control static vs. dynamic allocation and chunking
iii. `shared(...)`/`private(...)`: data-scoping clauses

Controlled:

i. `#pragma omp sections`/`#pragma omp section`: work-sharing over per-thread sections
ii. `#pragma omp single`: one thread executes block
iii. `#pragma omp master`: only master executes block

Synchronisation:

i. `#pragma omp barrier`: explicit barrier for threads
ii. `#pragma omp critical`: critical section block
iii. `#pragma omp atomic`: atomic update to a location

Mutex:

```
omp_lock_t lock;
omp_init_lock(&lock);
#pragma omp parallel
{
  omp_set_lock(&lock);
  x++;
  omp_unset_lock(&lock);
}
```

## NVIDIA CUDA

CUDA Programming Model:

i. Heterogeneous: Host (CPU) + Device (GPU)
ii. Kernel: function that runs on the device
iii. Thread: local registers and memory; `threadIdx`
iv. Warp: execution unit of 32 threads scheduled together (SIMT); one instruction at a time
v. SM (Streaming Multiprocessor): schedules warps with resources (register file, shared memory); thread blocks execute concurrently on an SM

Function Qualifiers:

| Qualifier | Scope | Usage |
|---|---|---|
| `__host__` | CPU | local to host |
| `__device__` | GPU | local to device |
| `__global__` | GPU | launch from host |

Variable Qualifiers:

| Qualifier | R/W | Cache | Scope | Lifetime |
|---|---|---|---|---|
| `__device__` | R/W | No | grid | static |
| `__constant__` | R | Yes | grid | static |
| `__shared__` | R/W | No | block | block |
| unqualified | R/W | – | thread | thread |
| `__managed__` | R/W | – | global | static |

Optimizations:

i. Reduce transfers: minimize host↔device copies; batch small copies; use pinned memory; overlap copy/compute with `cudeMemcpyAsync` + streams
ii. Global memory: coalesce accesses; minimize global loads/stores; use `__shared__`; avoid bank conflicts
iii. Occupancy: many warps to hide latency; threads/block multiple of 32; balance registers/shared-mem so $\geq 1$ block/SM
iv. Control flow: minimize warp divergence (avoid thread-ID dependent branches)
v. Compute: prefer high-throughput ops; single precision faster; avoid int div/mod (use shifts/bitwise)
vi. Avoid multi-context: do not create multiple CUDA contexts per GPU within one application

### Matrix Addition (Device)

```
__global__
void addMatrixG(float *a,float *b,float *c,int N)
{
  int i = blockIdx.x * blockDim.x + threadIdx.x;
  int j = blockIdx.y * blockDim.y + threadIdx.y;
  int index = i + j * N;
  if (i < N && j < N)
    c[index] = a[index] + b[index];
}
```

### Matrix Multiplication (Device)

```
__global__
void addMatrixG(float *a,float *b,float *c,int N)
{
  int i = blockIdx.x * blockDim.x + threadIdx.x;
  int j = blockIdx.y * blockDim.y + threadIdx.y;
  int index = i + j * N;
  if (i < N && j < N)
    c[index] = a[index] + b[index];
}
```

### Host Skeleton

```
void main(){
  // Assume host arrays: h_A, h_B, h_C; device
    pointers: d_A, d_B, d_C
  size_t bytes = (size_t)N * sizeof(float);
  cudaMalloc((void**)&d_A, bytes);
  cudaMalloc((void**)&d_B, bytes);
  cudaMalloc((void**)&d_C, bytes);

  // Copy inputs Host -> Device
  cudaMemcpy(d_A, h_A, bytes,
    cudaMemcpyHostToDevice);
  cudaMemcpy(d_B, h_B, bytes,
    cudaMemcpyHostToDevice);

  // Configure grid/block and launch kernel
  dim3 dimBlk(BLOCK_X, BLOCK_Y);    // e.g.
    (16,16)
  dim3 dimGrd((NX + BLOCK_X - 1)/BLOCK_X,
              (NY + BLOCK_Y - 1)/BLOCK_Y);
  kernel<<<dimGrd, dimBlk>>>(d_A, d_B, d_C, ...);

  // Copy result Device -> Host
  cudaMemcpy(h_C, d_C, bytes,
    cudaMemcpyDeviceToHost);

  // Free device memory
  cudaFree(d_A); cudaFree(d_B); cudaFree(d_C);
}
```

## MPI

Message Passing Model:

i. Distributed memory
ii. Explicitly represent parallelism with send/receive

MPI Program Overview:

i. Initialize communications
ii. Communicate to coordinate computation
iii. Exit from message-passing system

### Communication Semantics

Protocol Choice:

i. Buffered: sender returns after copying data into a communication buffer (less idling; buffer overhead)
ii. Non-buffered: sender blocks until matching receive is encountered (idling + deadlock risks if mismatch)

Local View:

i. Blocking: return ⇒ safe to reuse resources/buffers
ii. Non-blocking: may return before completion; programmer must ensure completion (poll/check)

Global View:

i. Synchronous: operation does not complete before both sides have started
ii. Asynchronous: sender can proceed without coordination with receiver

Operations:

i. Sync + Blocking: `MPI_Ssend`
ii. Async + Blocking: `MPI_Send` / `MPI_Recv`
iii. Sync + Non-blocking: `MPI_Issend`
iv. Async + Non-blocking: `MPI_Isend` / `MPI_Irecv`

* Note: blocking and non-blocking ops can be mixed

### Library Functions

Initi, Finalize, Abort Calls:

i. `int MPI_Init(int* argc, char** argv[])`
ii. `int MPI_Finalize(void)`
iii. `int MPI_Abort(MPI_Comm comm, int errorCode)`

Point-to-Point Messaging Calls:

i. `int MPI_Send(void* buf, int count, MPI_Datatype dt, int dst, int tag, MPI_Comm c)`
ii. `int MPI_Recv(void* buf, int count, MPI_Datatype dt, int src, int tag, MPI_Comm c, MPI_Status *status)`

Message Format:

i. Data: start-buffer + count + datatype
ii. Envelope: destination/source (rank) + tag + communicator
iii. Receive buffer must be ≥ message length
iv. Wildcards: `MPI_ANY_SOURCE`, `MPI_ANY_TAG`
v. `MPI_Status`: `MPI_SOURCE`, `MPI_TAG`, `MPI_ERROR`

Ordering:

i. 1 sender, 1 receiver: messages delivered in order
ii. > 2 processes: message delivery order undefined

### Deadlocks

Deadlock Types:

i. Message Order: both sides wait on receives
ii. Buffer: no/small buffers ⇒ sends cannot complete

Secure MPI program: correctness does not depend on assumptions about MPI runtime properties.

- Specify send/receive order (e.g. even: send→recv, odd: recv→send)

### Collective Communication

Process Groups are ordered sets of processes:

i. Unique rank identifier
ii. Processes may be in multiple groups
iii. Communicator: communication domain for a group
   - Intra-communicator: collectives in a group; default `MPI_COMM_WORLD`
   - Inter-communicator: P2P between two groups

Virtual Topology is a Cartesian-style communicator:

i. `MPI_Cart_create`, `MPI_Cart_get`, `MPI_Cartdim_get`, `MPI_Cart_coords`, `MPI_Cart_rank`, `MPI_Cart_shift`

Collectives involve all processes in a communicator:

i. Total Exchange: `MPI_Alltoall`
ii. Multi-broadcast: `MPI_Allgather`
iii. Multi-accumulation: `MPI_Reduce_scatter`
iv. Scatter/Gather: `MPI_Scatter` / `MPI_Gather`
v. Single Broadcast: `MPI_Broadcast`
vi. Single Accumulation: `MPI_Reduce`
vii. Barrier: `MPI_Barrier`

### Hello World (Master Sends)

```
MPI_Init(&argc, &argv);
MPI_Comm_size(MPI_COMM_WORLD, &size);
MPI_Comm_rank(MPI_COMM_WORLD, &rank);
tag = 100;
if (rank == 0) {
  strcpy(message, "Hello World 2");
  for (i=1; i<size; i++)
    MPI_Send(message, 14, MPI_CHAR, i, tag,
    MPI_COMM_WORLD);
} else {
  MPI_Recv(message, 14, MPI_CHAR, 0, tag,
    MPI_COMM_WORLD, &status);
}
printf("node %d : %.13s\n", rank, message);
MPI_Finalize();
```

# 4. Performance

Performance is measured by primary goals:

i. Response Time: wall-clock duration of an execution consists of user, system and waiting time
ii. Throughput: average work completed per unit time

## Sequential Programs

User CPU Time:

$$T_{user} = N_{cycle} \times T_{cycle}$$

i. $T_{cycle} = \frac{1}{\text{(average) clock freq.}}$
ii. $N_{cycle} = \sum n_i \times CPI_i \approx N_{instr} \times CPI$

User CPU Time (Memory Access; One-level Cache):

$$T_{user} = \left(N_{cycle} + N_{mm_cycle}\right) \times T_{cycle}$$

i. $N_{mm_cycle} = N_{read_cycle} + N_{write_cycle}$
ii. $N_{read_cycle} = N_{reads} \times R_{miss_rate} \times N_{miss_cycles}$

Average Memory Access Time (Two-level Cache):

$$T_{read} = T_{hit} + \left(R_{miss} \times T_{miss}\right)$$

i. $T_{read} = T_{hit}^{L1} + \left(R_{miss}^{L1} \times T_{miss}^{L1}\right)$
ii. $T_{miss}^{L1} = T_{hit}^{L2} + \left(R_{miss}^{L2} \times T_{miss}^{L2}\right)$
iii. Global miss rate: $R_{miss} = R_{miss}^{L1} \times R_{miss}^{L2}$

User CPU Time (Integrated CPI + Memory):

$$T_{user} = \Big(N_{cycle} + [N_{rw_ops} \times R_{miss} \times N_{miss_cycles}]\Big) \times T_{cycle}$$

Throughput Metrics:

i. MIPS $= \frac{N_{instr}}{T_{user} \times 10^6}$
ii. MFLOPS $= \frac{N_{fl_ops}}{T_{user} \times 10^6}$
   + Restrict to float ops (but variance exists)
iii. Arithmetic Intensity (AI) $= \frac{N_{fl_ops}}{N_{bytes_rw}}$
   + Roofline: $P \leq \min\left(P_{\text{peak}},\ B \cdot \text{AI}\right)$ shows if bound by computation or communication bandwidth

## Parallel Programs

Parallel Execution Time on $p$ PUs, $T_p$ consists:

i. Time for executing local computations
ii. Time for exchange of data between PUs
iii. Time for synchronization between PUs
iv. Waiting time

Speedup:

$$S_p = \frac{T_*}{T_p}$$

i. Measures the benefit of parallelism compared to best sequential algorithm time $T_*$
ii. Theoretically, $S_p \leq p$ always holds
iii. In practice, $S_p > p$ (superlinear speedup) can occur (e.g. problem working set "fits" in the cache)

Cost:

$$C_p = p \times T_p$$

i. Measures total work performed by all PUs i.e. processor-runtime product
ii. Cost-optimal: parallel program executes same total number of operations as fastest sequential program

Efficiency:

$$E_p = \frac{T_*}{C_p} = \frac{S_p}{p} = \frac{T_*}{p \times T_p}$$

i. Measures actual degree of speedup performance achieved compared to maximum
ii. Theoretically, ideal $S_p = p \ \Rightarrow\ E_p = 1$
iii. In practice, $E_p > 1$ can occur with superlinear speedups

## Scalability

Scalability is the interaction between problem size and parallel machine size:

i. Impacts: load balancing, overhead, arithmetic intensity, locality (application dependent)
ii. Fixed problem size:
   - Small $n$: overheads dominate $\Rightarrow$ suitable for small machine, poor on large machine
   - Large $n$: working set may not fit on small machine (thrashing / exceeds cache / can't run)
iii. Constraints:
   - Application: particles/processor (N-body), transactions/processor (distributed DB)
   - Resources: time-limited, memory-limited, problem-limited (same problem faster)

Amdahl's Law:

$$S_p(n) = \frac{T_*(n)}{f \cdot T_*(n) + \frac{1-f}{p} T_*(n)} = \frac{1}{f + \frac{1-f}{p}} \leq \frac{1}{f}$$

i. Sequential fraction: $0 \leq f \leq 1$ limits speedup
ii. Discourage from making large parallel computers, focus on reducing sequential fraction $f$
iii. Rebuttal: circumvented for large problem size
   - Commonly $f$ is a function of $n$, $f(n)$
   - Efficient parallel algorithm: $\lim_{n\to\infty} f(n) = 0$
   - Thus, $\lim_{n\to\infty} S_p = \frac{p}{1+(p-1)f(n)} = p$

Gustafson's Law:

$$S_p(n) = \frac{\tau_f + \tau_v(n,1)}{\tau_f + \tau_v(n,p)} = f + (1-f)p$$

i. Encouraged to use more processors to solve larger problem in the same time (in CPU-bound apps)
ii. $\tau_f$: constant sequential execution time
iii. $\tau_v(n,p)$: parallel execution time on $p$ processors; if perfectly parallelizable (no overheads):
   - $\tau_v(n,1) = T^*(n) - \tau_f$
   - $\tau_v(n,p) = \frac{T^*(n) - \tau_f}{p}$
iv. If $T^*(n)$ increases strongly monotonically with $n$, then $\lim_{n\to\infty} S_p(n) = p$

## Communication

Communication Performance Measures:

i. Bandwidth, $B$: maximum data rate (bytes/s)
ii. Throughput: effective bandwidth (bytes/s)
iii. Time of Flight, $T_{delay}$: time for first bit to arrive
iv. Byte Transfer Time, $t_B$: time to transmit one byte
v. Transmission Time: time to transmit message
vi. Transport Latency: time to transfer a message (time of flight + transmission time)
vii. Sender Overhead, $O_{send}$: time to compute checksum, append header, and execute routing
viii. Receiver Overhead, $O_{recv}$: time to compare checksum and generate acknowledgement

Communication Time (message size $m$):

$$\begin{aligned} T(m) &= O_{send} + T_{delay} + \frac{m}{B} + O_{recv} \\ &= T_{overhead} + \frac{m}{B} \\ &= T_{overhead} + t_B \cdot m \end{aligned}$$

where:

i. $T_{overhead} = O_{send} + T_{delay} + O_{recv}$ is independent of message size $m$
ii. Byte Transfer Time, $t_B = \frac{1}{B}$
iii. Assume no checksum error, and no network contention / congestion

## Performance Analysis

Practical Workflow:

i. Measure first; define target metric (latency vs throughput vs speedup)
ii. Localise the bottleneck (CPU vs memory vs sync vs I/O vs network)
iii. Change *one* factor at a time; keep workload and environment fixed
iv. Validate: speedup should be explained by reduced dominant component

## Instrumentation

Performance Instrumentation finds performance metrics:

i. Latency: time to service a request
ii. Response Time: wall-clock duration of an execution
iii. Throughput: average work completed per unit time
iv. IOPS: I/O operations per second
v. Utilization: fraction of time a resource is busy
vi. Saturation: degree to which a resource has queued work it cannot service

Perspectives:

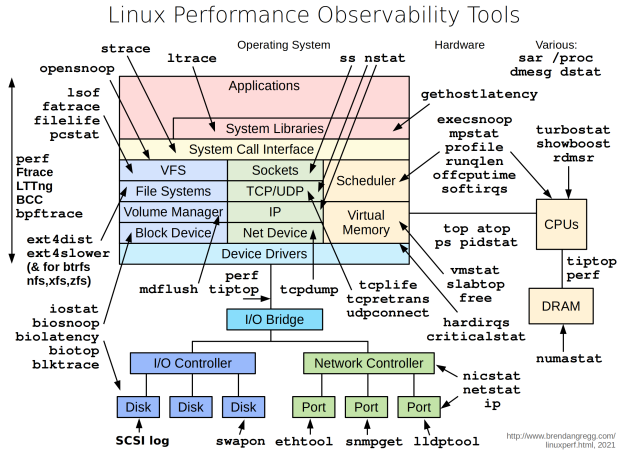
i. Resource Analysis: focus on utilization
   - Time-based: average time the resource was busy
   - Capacity-based: resource throughput
ii. Workload Analysis: focus on throughput, latency

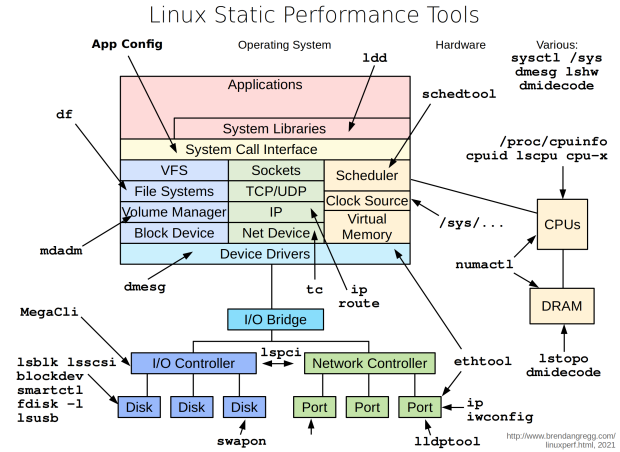
Methodologies:

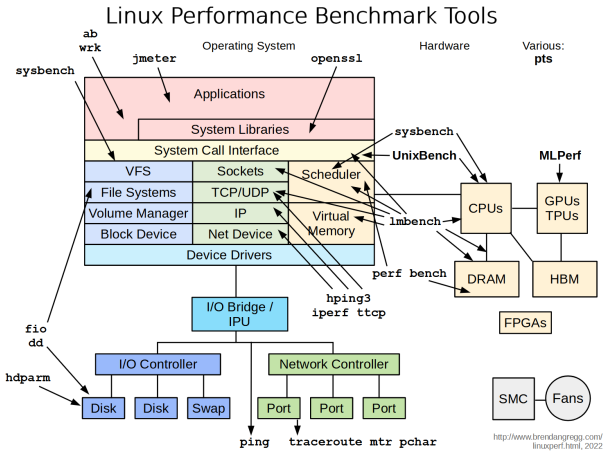
i. Anti-Methodologies: undeliberate, street light (look for obvious issues), drunk man (tune at random)
ii. Problem Statement
iii. USE (per resource): Utilization, Saturation, Errors
iv. Monitoring: record metrics over time

Tool Types & Counters:

i. Observability: watch activity under workload (timing statements, performance counters)
ii. Static: examine system at rest
iii. Benchmarking: load test (caution: contention/production risk)
iv. Tuning: change defaults (danger: regressions now/later under load)
v. Categorization:
   - System-wide vs per-process
   - Fixed counters (kernel-maintained metrics) vs event-based counters (enabled as needed)
   - Profiling: samples/snapshots vs Tracing: record every event occurrence



## 5. Energy Efficiency

Energy-Efficiency Motivation:

i. Power dissipation has become a limiting factor (*"power wall"*)
ii. Higher performance ⇒ more/faster computers ⇒ more power/heat/cooling/space/cost

### Mobile Computing

Trends:

i. Energy-efficiency: decrease power consumption in hardware and increase performance
ii. Increase battery capacity
iii. ARM processing systems:
   - Designed for power-efficiency
   - Closing in on x64 performance
   - Highly customizable architecture

ARM big.LITTLE (low-power approach):

i. Big CPU: high performance for intensive workloads
ii. Little CPU: low power for majority workloads
iii. Switch between cores depending on demand

### Heterogeneous Computing

Heterogeneous Platforms:

i. Processors: brawny vs wimpy, big–little
ii. Supercomputers: accelerators
iii. Data Centers: mixed server generations
iv. Cloud: heterogeneous resources with different price–performance

Challenges:

i. Use heterogeneity to reduce power while maintaining performance
ii. Energy-efficient configuration for parallel applications is complex (scheduling)
iii. Burden shifts to programmers for portable code

Data Centers:

i. Large-scale data centers consume significant electricity and emit substantial $CO_2$
ii. Cooling overhead can be comparable to computation energy
iii. Power Use Efficiency (PUE) = $\frac{\text{Total Energy}}{\text{IT Energy}}$: measures energy efficiency in data centers

Google (energy-efficiency practices):

i. Continuously measure efficiency:
   - Computation (IT) energy
   - Overhead: cooling and conversions
ii. Build custom, highly-efficient servers:
   - Minimize AC/DC conversion losses
   - Remove unnecessary parts
   - Strategic rack placement
   - Tune fan speeds for cooling
iii. Extend equipment lifecycle
   - Reuse, resell components
iv. Control equipment temperature:
   - Raise operating temperature (e.g. 26°C)
   - Thermal modeling
v. Cooling with water instead of chillers:
   - Evaporative cooling, sea water, recycled water

Reducing Energy Consumption:

i. Move less data:
   - Reduce transfers to/from memory
   - Exploit locality
   - Use compression
ii. Use specialized processing:
   - Compute less (no parallel if it increases work)
   - CPU-like cores + GPU-like throughput-optimized cores
   - FPGAs (programmable hardware)

## 6. Appendix

| name | prefix | multiplier |
|---|---|---|
| exa | E | $10^{18}$ |
| peta | P | $10^{15}$ |
| tera | T | $10^{12}$ |
| giga | G | $10^{9}$ |
| mega | M | $10^{6}$ |
| kilo | K | $10^{3}$ |
| milli | m | $10^{-3}$ |
| micro | $\mu$ | $10^{-6}$ |
| nano | n | $10^{-9}$ |
| pico | p | $10^{-12}$ |

| name | prefix | multiplier |
|---|---|---|
| exbi | Ei | $2^{60}$ |
| pebi | Pi | $2^{50}$ |
| tebi | Ti | $2^{40}$ |
| gibi | Gi | $2^{30}$ |
| mebi | Mi | $2^{20}$ |
| kibi | Ki | $2^{10}$ |